



兰州工业学院

实 验 报 告

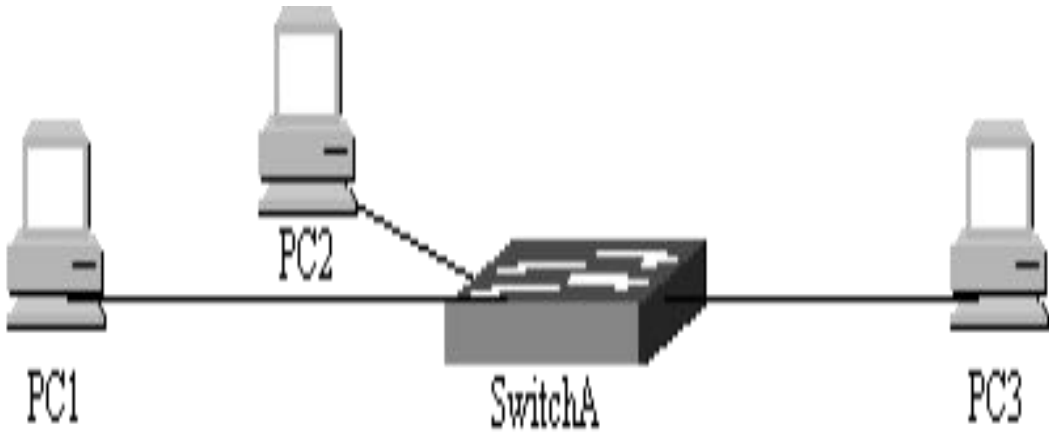
(2023—2024 学年第 2 学期)

课程名称 《计算机网络应用》

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实验项目	实验二 VLAN 间通信		
实验时间	2024 年 4 月 8 日	实验地点	信息与通信工程基础实验室 (EDA 技术实验室)
一、实验目的 1. 理解虚拟局域网的功能与作用； 2. 掌握交换机的基本配置命令； 3. 基于端口的 VLAN 划分方法； 4. 掌握使用三层交换机实现不同 VLAN 间的通信。 二、实验环境 Cisco Packet Tracer 模拟软件 三、实验步骤 1、 在一台 2960 交换机上设置 VLAN 任务要求： (1) 在一台已经进行初始设置的 2960 交换机上设置两个 VLAN：VLAN10、VLAN20，分别取名为 students、teachers。 (2) PC1 和 PC2 是 students 中的成员，PC3 是 teachers 中的成员。 (3) 验证设置前后 PC1、PC2、PC3 是否能够 ping 通。  The diagram shows a network topology with three PCs (PC1, PC2, PC3) and a central switch labeled 'SwitchA'. PC1 and PC2 are connected to the left side of the switch, while PC3 is connected to the right side. All connections are represented by straight lines.			
图 1 实验拓扑结构图			

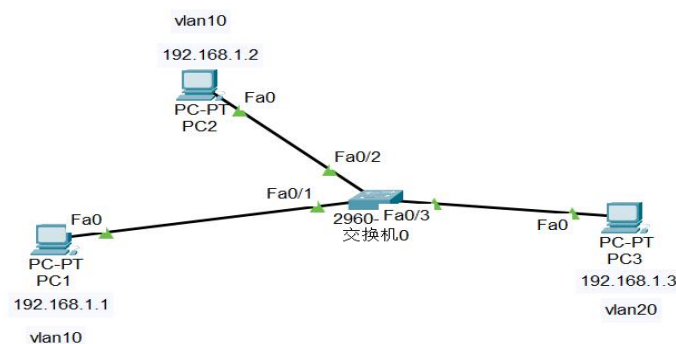


图 1 在 Cisco Packet Tracer 模拟软件的拓扑图

```

!
interface FastEthernet0/1
 switchport access vlan 10
!
interface FastEthernet0/2
 switchport access vlan 10
!
interface FastEthernet0/3
 switchport access vlan 20
!

```

图 2 PC1、PC2、PC3 和交换机连接的端口信息

PC1 192.168.1.1, 255.255.255.0

PC2 192.168.1.2, 255.255.255.0

PC3 192.168.1.3, 255.255.255.0

图 3 PC1、PC2、PC3 的 IP 地址、子网掩码及网关信息

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

图 4 PC1 ping192.168.1.2 和 1.3 发现能够 ping 通

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<lms TTL=128
Reply from 192.168.1.1: bytes=32 time<lms TTL=128
Reply from 192.168.1.1: bytes=32 time<lms TTL=128
Reply from 192.168.1.1: bytes=32 time<lms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<lms TTL=128
Reply from 192.168.1.3: bytes=32 time<lms TTL=128
Reply from 192.168.1.3: bytes=32 time<lms TTL=128
Reply from 192.168.1.3: bytes=32 time<lms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

图 5 PC2 ping 192.168.1.1 和 1.3 发现能够 ping 通

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<lms TTL=128
Reply from 192.168.1.1: bytes=32 time<lms TTL=128
Reply from 192.168.1.1: bytes=32 time<lms TTL=128
Reply from 192.168.1.1: bytes=32 time<lms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|

```

图 6 PC3 ping 192.168.1.1 和 1.2 能够

```

>switch>ena
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name students
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name teachers
Switch(config-vlan)#exit
Switch(config)#inte
Switch(config)#interface fa
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switch
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#switchport access vlan 20
Switch(config-if)#

```

图 7 在交换机配置 vlan，设置了 vlan10 和 vlan20 两个 vlan 口，分别命名 students、teachers，将连接的 f 端口将入对应的 vlan 口

```

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

图 8 配置 vlan 之后发现 PC1 能够 ping 通同一 vlan 口的 PC2，不同 valn 口的 PC3ping 通不了

```

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

图 9 配置 vlan 之后发现 PC2 能够 ping 通同一 vlan 口的 PC1，不同 valn 口的 PC3ping 通不了

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

图 10 发现和其他 vlan 口的两个 PC ping 通不了

2、在一台 3560 三层交换机上实现不同 VLAN 间的通信。

任务要求：

- (1) 在一台已经进行初始设置的 2960 交换机上设置三个 VLAN：VLAN10、VLAN20、VLAN30。
- (2) PC0 是 VLAN10 中的成员，PC1 是 VLAN20 中的成员，PC2 是 VLAN30 中的成员。
- (3) 验证设置前后 PC1、PC2、PC3 是否能够 ping 通。

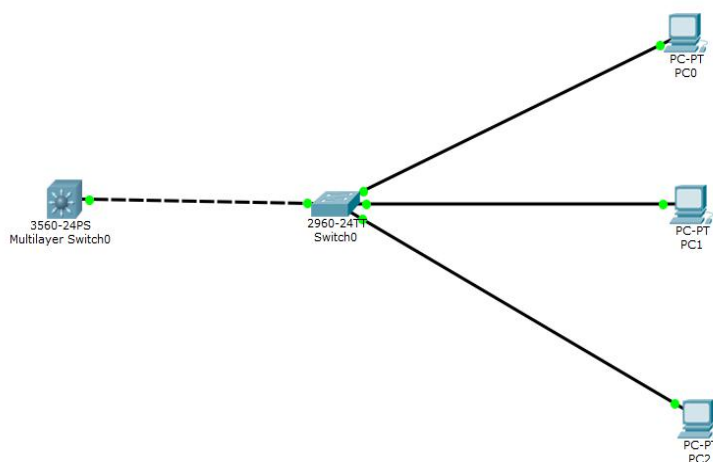


图 1 实验拓扑结构图

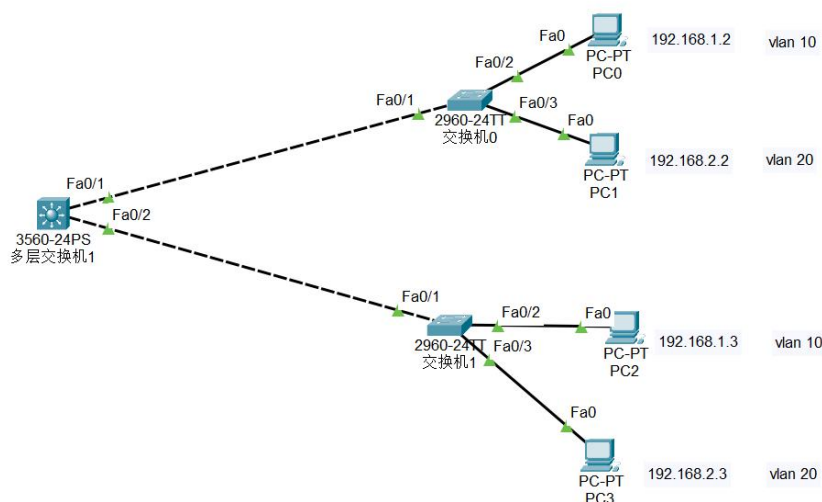


图 11 在 Cisco Packet Tracer 模拟软件的拓扑图

```

interface FastEthernet0/2
switchport access vlan 10
!
interface FastEthernet0/3
switchport access vlan 20
!

```

图 12 PC1、PC2、PC3 与交换机 2960 连接的端口信息

```

interface FastEthernet0/1
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk encapsulation dot1q
switchport mode trunk
!

```

图 13 交换机 2960 与交换机 3560 连接的端口信息

```

PC0:192.168.1.2, 255.255.255.0
PC2:192.168.1.3, 255.255.255.0
网关: 192.168.1.1

PC1:192.168.2.2, 255.255.255.0
PC3:192.168.2.3, 255.255.255.0
网关: 192.168.2.1

```

图 14 PC1、PC2、PC3 的 IP 地址、子网掩码及网关信息

```

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<lms TTL=128
Reply from 192.168.1.3: bytes=32 time<lms TTL=128
Reply from 192.168.1.3: bytes=32 time<lms TTL=128
Reply from 192.168.1.3: bytes=32 time<lms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

图 15 PC0 ping 192.168.1.3, 192.168.2.2 发现同一网段的能够 ping 通，不同网段的无法 ping 通

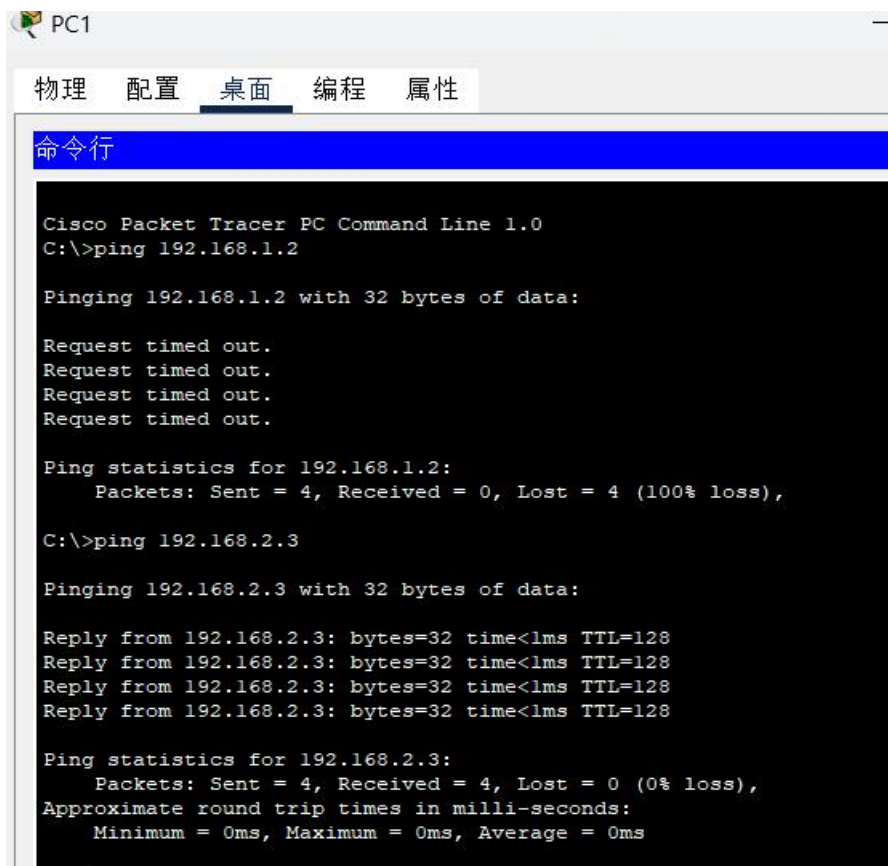
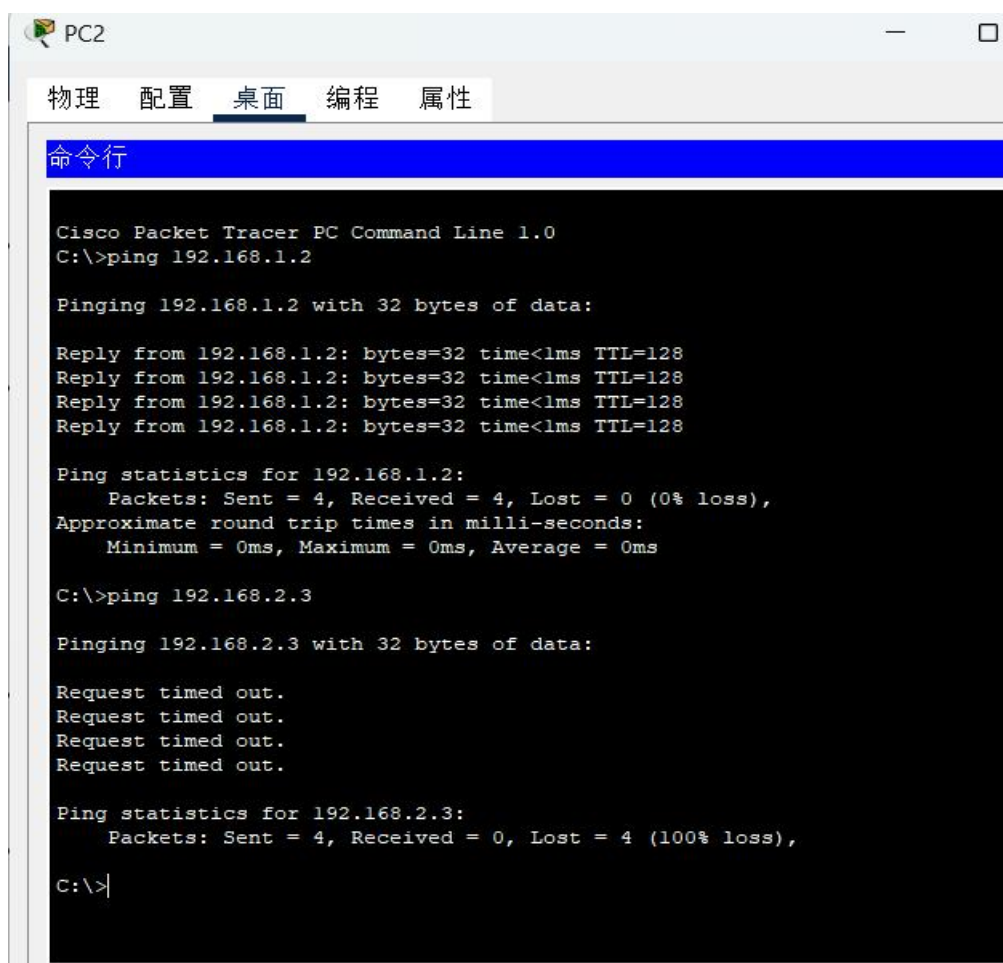


图 16 PC1 ping 192.168.1.2, 192.168.2.3 发现同一网段的能够 ping 通，不同网段的无法 ping 通



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC2. The window has tabs for '物理' (Physical), '配置' (Configuration), '桌面' (Desktop), '编程' (Programming), and '属性' (Properties). The '配置' tab is active, and the '命令行' (Command Line) section is highlighted. The command prompt shows the following sequence of commands and outputs:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

图 17 PC2 ping 192.168.1.2, 192.168.2.3 发现同一网段的能够 ping 通，不同网段的无法 ping 通

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#inter
Switch(config)#interface fa0/2
Switch(config-if)#switc
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#intre
Switch(config)#inter
Switch(config)#interface fa0/1
Switch(config-if)#switc
Switch(config-if)#switchport mode tru
Switch(config-if)#switchport mode trunk
```

图 18 进入全局配置模式，创建了 vlan10、vlan20 两个 vlan，进入连接 PC 机的端口，将该端口分配到对应的 vlan，进入连接三层交换机的端口，将端口设置为 trunk 模式

```

Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#inter
Switch(config)#interface fa0/1
Switch(config-if)#switc
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk en
Switch(config-if)#switchport trunk encapsulation do
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#swit
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit

```

图 19 进入全局配置模式，创建了 vlan10、vlan20 两个 vlan，然后进入连接二层交换机的端口，将 trunk 封装设置为 802.1q 协议，然后将端口设置为主干道 Trunk 模式

```

Switch(config)#inter
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

Switch(config-if)#ip address 192.168.1.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

Switch(config-if)#ip address 192.168.2.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#ip routing
Switch(config)#

```

图 20 进入 vlan 的配置模式，为 vlan 接口配置 IP 地址和子网掩码，然后开启该端口，最后使用 ip routing 命令可以开启三层交换机的 IP 路由功能。

```

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

图 21 PC0 ping192.168.2.2 和 192.168.2.3 发现都能 ping 通

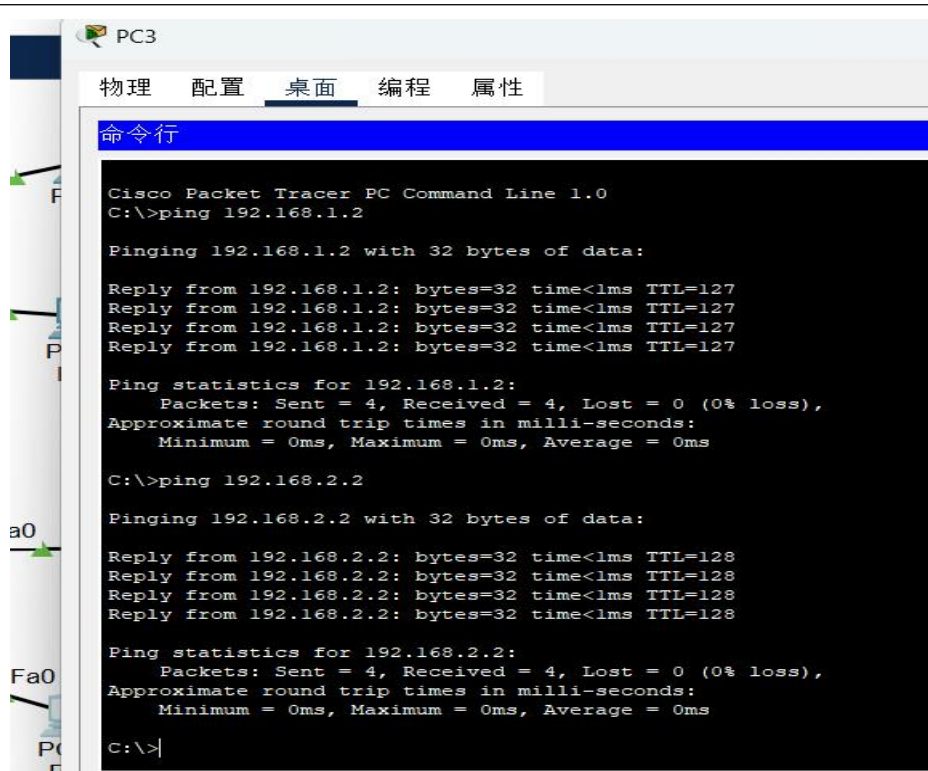


图 22 ping192.168.1.2 和 192.168.2.2 也能够 ping 通

四、总结

实验成绩评定表

序号	考核项目	分值分布	成绩
1	出勤与纪律情况	10	
2	实验任务完成情况	40	
3	实验报告质量	50	
总分			
指导教师签字			

备注：考核项目及分值分布应依据教学大纲确定。